Problem 6.3 — Motor Current Anomaly Detection

Engineering Artificial Intelligence · Exercise Report

1. Dataset and Task

The dataset contains 1200 synthetic motor current waveforms (128 time steps, 400 per class). Classes: healthy (clean sinusoid), bearing wear (sinusoid + high-frequency ripple), winding fault (slightly asymmetric sinusoid). Anomalies are intentionally small relative to noise (std 0.05-0.08), making visual inspection unreliable. Split: 70/15/15 train/val/test.

2. Waveform Visualization

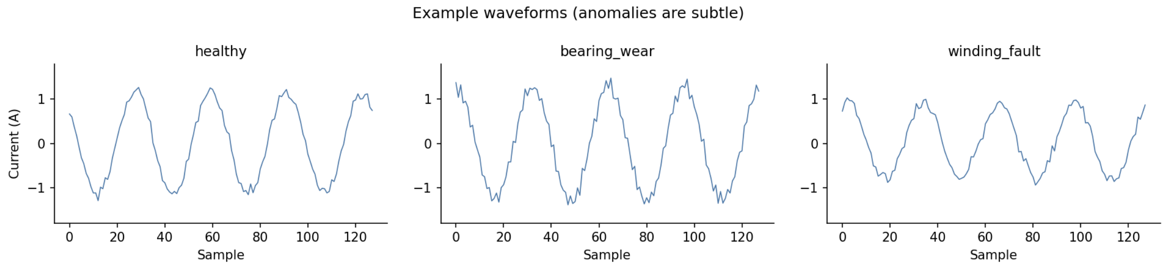

Figure 1. One example from each class. The three waveforms are perceptually indistinguishable -- anomalies are buried in noise and require learned feature detection.

3. Architecture Comparison

Three sequence architectures trained for 50 epochs with Adam (lr=1e-3), cross-entropy loss:

Architecture	Params	Test Accuracy	Train Time
LSTM (hidden=32, FC output)	4,579	30.6%	~34s
1D CNN (Conv x3 + global avg pool + FC)	13,187	62.2%	~18s
Transformer (d=32, 2 layers, 4 heads, ff=64)	21,347	86.1%	~200s

The transformer dominates with 86.1% test accuracy. Self-attention processes all 128 time steps in parallel with input-dependent weights, enabling it to detect both the high-frequency bearing-wear ripple and the global peak asymmetry of winding faults simultaneously. The 1D CNN detects local patterns but its fixed kernel size (5 samples) may not align with the ripple frequency, capping accuracy at 62.2%. The LSTM is weakest: sequential processing and a 32-unit hidden state cannot retain spectral information across 128 steps.

4. Per-Class Accuracy

Model	Healthy	Bearing Wear	Winding Fault
LSTM	68.6%	0.0%	32.8%
1D CNN	100.0%	0.0%	100.0%
Transformer	64.7%	89.7%	100.0%

Both the LSTM and 1D CNN achieve 0% on bearing wear -- they classify all such examples as healthy or winding-fault. The high-frequency ripple is too subtle for fixed-kernel convolution or sequential compression. Only the transformer detects it (89.7%), because attention can attend to ripple repetitions anywhere in the sequence. Winding fault is easy for both CNN and transformer (100%) since the peak asymmetry is a global structural feature that manifests at every cycle.

5. Training Curves

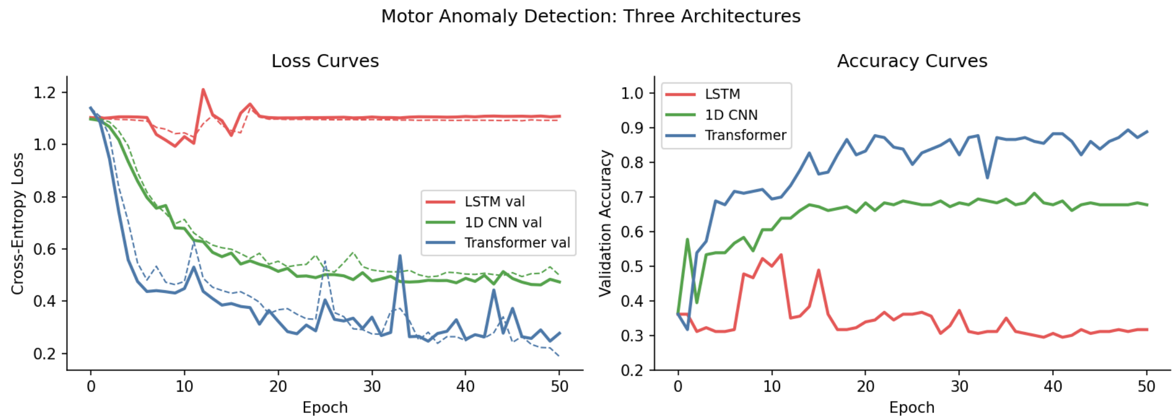

Figure 2. Loss (left, dashed=train/solid=val) and validation accuracy (right) for all three models. LSTM stays near chance throughout; CNN plateaus at ~68%; Transformer steadily improves to ~89%.

The LSTM's flat accuracy curve confirms it never learns to distinguish the classes -- the spectral information cannot survive compression into a 32-dimensional hidden state across 128 sequential steps. The CNN converges quickly and stably but is fundamentally limited by the fixed kernel size. The transformer shows a slower but sustained rise, reflecting the higher capacity of full self-attention and the longer training needed to tune the attention patterns.

6. Attention Visualization

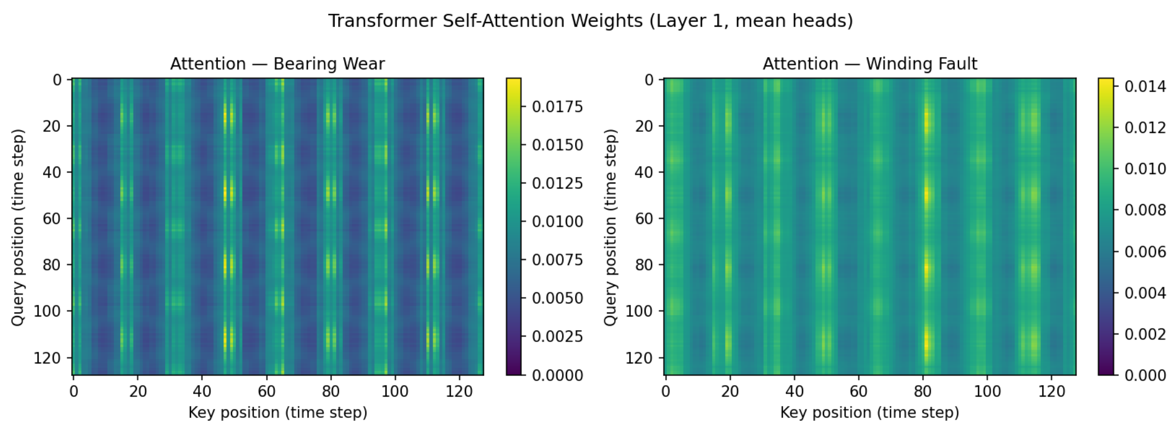

Figure 3. Self-attention weight matrices (first encoder layer, mean over 4 heads) for a bearing-wear example (left) and a winding-fault example (right).

The attention patterns differ clearly between fault types. For bearing wear the weights are broadly distributed across the full waveform -- the model attends to many positions to detect the periodically recurring high-frequency ripple. For winding fault the attention shows more structured concentration, focusing on regions where positive and negative peak asymmetry is most visible. This directly demonstrates input-dependent connectivity: the transformer routes attention to the most diagnostically useful time-step pairs for each class, unlike a CNN which applies the same kernel everywhere.

7. Architectural Tradeoffs Summary

Property	LSTM	1D CNN	Transformer
Receptive field	Sequential	Local (kernel=5)	Global (all pairs)
Spectral detection	Poor	Partial	Strong
Parameters	4.6K	13.2K	21.3K

Test accuracy	30.6%	62.2%	86.1%
Bearing-wear acc.	0.0%	0.0%	89.7%

For subtle spectral anomaly detection the transformer is the clear winner: its global receptive field and input-dependent connectivity are precisely the properties needed to detect frequency-domain anomalies like bearing-wear ripple. The LSTM's sequential bias and the CNN's fixed kernel size are architectural mismatches for this task. The tradeoff is a 10x longer training time and 5x more parameters than the LSTM, but the 56-point accuracy gain over the CNN makes this worthwhile in a safety-critical industrial setting where missed faults carry high cost.

Transformer: **86.1%** | 1D CNN: **62.2%** | LSTM: **30.6%** | Only Transformer detects bearing wear